Ethan Wu

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Profile

Dedicated and analytical 3rd-year Mechatronics and Robotics Engineering student at Queen's University with a strong foundation in embedded systems, high-level programming, CAD, and automated workflow design. Proven leadership experience managing the Automation Teams, developing robotics software stacks (C++, ROS2) for competitive design challenges, and coordinating multidisciplinary university projects.

Education

Queen's University Kingston, ON
Bachelor of Applied Science in Mechatronics and Robotics Engineering (GPA: 3.89) *Sept 2024 – Expected May 2028*
- Relevant Coursework: Sensors and Electric Actuators, CAD, Electronics, Signals and Systems, Microprocessor Interfacing, Embedded Systems, Computer Architecture, Feedback Control Systems, Manufacturing Processes, Corporate Law.

University of Toronto Schools (UTS) Toronto, ON
High School Diploma (Grade Average: 97%) *Sept 2018 – June 2024*
- Awards: AP Scholar Award, Ontario Scholar, Honor Roll (2019–2024).

Leadership & Extracurricular Experience

Task Subteam Lead – NASA HERC RC Challenge Sept 2026 – Present
Queen's Lunar Exploration Team (QLET) *Kingston, ON*
- Lead multidisciplinary engineering team for the 2027 NASA HERC competition, owning end-to-end design and execution of the autonomous task unit mechanism.
- Architecting the task execution state machine in ROS2 and C++ to parse incoming radio command sequences and trigger automated rover routines.
- Integrating a pan-tilt camera mechanism with OpenCV target recognition algorithms to isolate geometric shapes, calculate target distances via focal geometry, and dynamically apply digital color filters.
- Collaborating across mechanical and electrical subteams to prototype pan-tilt actuators, integrate optical sensors, and validate system power distribution.

Automation Team Manager (formerly Project Lead) Oct 2025 – Present
Engineering Society at Queen's University *Kingston, ON*
- Direct recruitment, developer onboarding, and project management activities for the Engineering Society automation team (approximately 20+ members).
- Created Power Automate flows to track training completions and streamline hiring workflows; collaborated with the Director of HR and VPFA to design and optimize future hiring protocols.
- Managing projects requested by Clark Hall Pub (real-time ordering queue), Campus Equipment Outfitters (dynamic merchandise ordering website), Director of Design Teams (an engineering extracurricular catalogue), etc..

Finance Coordinator Jul 2026 – Present
Queen's Space Conference *Kingston, ON*
- Managed financial logistics, budgeting, and sponsorship coordination for the annual space conference.

Challenges Coordinator Jun 2026 – Present
Queen's Engineering Competition *Kingston, ON*
- Designed, drafted, and budgeted the Senior Design Challenge for QEC 2026, establishing technical evaluation criteria.
- Validated proper electronic and mechanical materials via testing and prototyping to teams to ensure challenge feasibility in a 6-hour hackathon-style competition.

Embedded Systems General Member Sept 2025 – Jun 2026
Queen's AutoDrive Team *Kingston, ON*
- Engineered a C++ memory monitoring module for an embedded watchdog to detect real-time system crashes and memory leaks.
- Earned 2nd Place in the Mobility Innovation Challenge at the SAE AutoDrive Competition for an AV adoption framework.
- Earned 3rd Place Overall and 3rd Place in the Localization Challenge for AutoDrive Year 5.

Avionics General Member Sept 2025 – Apr 2026
Queen's Rocket Engineering Team (QRET) *Kingston, ON*
- Contributed to avionics circuit design, sensor integration, and telemetry systems for high-power rocketry.

QSuccess Mentor Jun 2025 – Apr 2026
Queen's University *Kingston, ON (Remote)*

- Supported incoming students throughout their first academic year, assisting mentees in establishing academic goals and fostering a growth mindset through bi-weekly mentoring sessions.

First-Year Project Coordinator Oct 2024 – Apr 2025
Queen's Rocket Engineering Team (QRET) *Kingston, ON*
- Drafted newsletters for sponsors and alumni; reviewed sponsorship and bursary documents; researched financial initiatives.

Engineering Orientation Leader (ACTION) Mar 2025 – Sep 2025
Queen's University Faculty of Engineering *Kingston, ON*
- Welcomed and mentored incoming first-year engineering students, creating an energetic, supportive environment and leading large-scale campus events.

VEX Robotics Team Senior Executive Sept 2023 – Jun 2024
University of Toronto Schools (UTS) *Toronto, ON*
- Trained 50+ members on robot design, CAD (SolidWorks/Fusion 360), prototyping, and mechanical/electrical assembly.
- Helped lead the team and delegate tasks at regional and international championships, including the VEX Robotics World Championship 2024.

Stage Crew Senior Executive & Member Sept 2021 – Jun 2024
University of Toronto Schools (UTS) *Toronto, ON*
- Elected executive member managing a 50+ member crew; mentored junior members on operating lighting and sound technology.
- Operated audio, lighting, and visual tech during large-scale live performances under strict time constraints.

Engineering Projects

TriPAD - Construction Site Sensor | *C++, ESP32, Sensor Fusion* Apr. 2026 – Aug. 2026
- Built a standalone environmental monitoring unit using an ESP32 to process particulate (PMS5003), noise (MAX9814), and IMU vibration (MPU6050) sensor telemetry over I2C and UART.
- Designed a 3D-printed Autodesk Fusion enclosure and secured 3rd place in the General's Lab Innovation Project (GLIP) competition.

Mars Sample Return (MSR) Rover | *ROS2, C++, Python, R&D* Jan. 2026 – Apr. 2026
- Developed a ROS2-based autonomous rover with a bidirectional Raspberry Pi-to-Arduino communication bridge.
- Implemented closed-loop PI control with IMU/encoder sensor fusion and integrated RPLIDAR A1 for Nav2 localization.

Prototype BB Ball Launcher Turret | *C++, Arduino, Autodesk Fusion, Motor Drivers* Oct 2026
- Designed and built a high-velocity pan-tilt BB launching turret utilizing DC motors, SG90/MG996R servos, DFRobot IO Expansion Shield v5, and DF-MD v1.3.
- Programmed Arduino Uno in C++ for closed-loop motor positioning and sensor-feedback control routines.
- Engineered custom structural mounting using everyday materials and 3D-printed parts in Autodesk Fusion.

Autonomous Transforming Robot (In-Progress) | *Autodesk Fusion, Kinematics, CAD* Oct 2026
- Designing a 30+ DOF kinematic assembly with interference checks, motion analysis, and tolerance tuning for 3D printing.

AI in Criminal Justice Research Paper | *UTS Capstone Project* 2023 – 2024
- Investigated AI integration in criminal justice, analyzing ethical implications regarding privacy, accuracy, and discrimination.
- Conducted survey-based research assessing adolescent perceptions of algorithmic justice to inform stakeholders.

Professional & Employment Experience

Teaching Assistant – APSC 101 Aug 2026 – Present
Stephen J. R. Smith Faculty of Engineering and Applied Science, Queen's University *Kingston, ON*
- Providing in-class academic support, troubleshooting, and design guidance for first-year engineering students in "APSC 101: Problem Analysis and Modeling."

Line Cook Jun 2025 – Aug 2025
Pearl Diver *Toronto, ON*
- Executed fast-paced kitchen operations, maintaining food safety standards and working efficiently under pressure in a high-volume restaurant environment.

Document Clerk May 2024 – Aug 2024
Houghton, Sloniowski & Stengel (formerly Youthfully) *Toronto, ON*
- Assisted with the preparation of numerous family law documents (including Applications, Affidavits, and Financial Statements) in compliance with legislation.

- Conducted legal case research and drafted internal legal memos to assist with active client files, sharpening extreme attention to detail.

Kitchen Assistant / Summer Camp Counselor Jun 2019 – Aug 2023
The Chef Upstairs *Vaughan, ON*
- Supervised and delegated tasks to campers aged 6–17 in cooking, cleanup, and workstation coordination, consistently delivering meals on schedule.
- Handled menu-planning, kitchen/pantry organization, inventory control, sanitation, and provided first aid when needed.

Interviewer and Surveyor 2019 – 2021
Angel Alert *Toronto, ON*
- Conducted structured interviews with elderly community members regarding personal safety, technology adoption, and social media habits.

Leaflet Distributor 2018 – 2020
Ontario Clean Air Alliance *Toronto, ON*
- Strategically planned optimal door-to-door distribution routes, completing 3+ hours of active fieldwork daily and delivering literature to over 1,200 households.

Athletics & Community Involvement

Soccer Player (Goalie & Defense) 2011 – 2018
Milliken Soccer Club *Markham, ON*
- Progressed from recreational level through house league and competitive rep teams over an 8-year span.

Certifications

- **Standard First Aid & CPR** – University of Toronto Schools (2024 – 2027)
- **Smart Serve** – Smart Serve Ontario (2025 – 2030)
- **WHMIS Certification**

Technical Skills

Software & Controls: C/C++, Python, Java, Reinforcement Learning (RL), ROS2, Nav2, SLAM, OpenCV, PID/PI Control, Sensor Fusion, Kinematics, SQL, Linux, HTML, CSS
Hardware & Embedded: ESP32, Raspberry Pi, Arduino, VHDL, CAN bus, UART, I2C, PMS5003, MAX9814, MPU6050, LiDAR, Encoders
CAD & Tools: Autodesk Fusion, SOLIDWORKS, VHDL, LTSpice, Quartus, MATLAB, MS Power Automate, Git, LaTeX
Hands-On: Soldering, Circuit Debugging, Machining, Welding, Woodworking, 3D Printing
Languages: English, Cantonese, Mandarin, French